

Proposal for Machine Learning Operations Project

Semester: HS26

Project Title	Urban Canopy Change Forecast: Prioritizing Tree Replacement in Zürich
Course / Module	MLOPS H26
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1 Short Project Description

The project estimates how crown-covered area changes in each 100 m cell of Zürich between LiDAR epochs. It presents those estimates as a prioritization map showing where canopy loss is likely and helping Grün Stadt Zürich identify areas for replacement planting.

2 Problem Statement

Problem: Zürich aims to increase crown cover in its built-up area from 15% (2022) to 25% by 2050. LiDAR measurements show a 64 ha decline in crown area from 2018 to 2022 (Stadt Zürich, 2023). Replacement trees take time to provide equivalent canopy, and planting budgets are limited, so the city needs to know where losses are concentrated.

Prediction: predict the change in crown-covered area, in m², for each 100 m cell between LiDAR epochs. Training uses the measured 2017/18–2021/22 change; for the prototype, 2021/22 features are used to estimate change to the next epoch. Predicted changes are grouped into map categories, such as likely loss, little change, or gain.

Horizon: the next LiDAR epoch, expected roughly four years later; the map can support annual replacement planning.

Scope: City of Zürich, non-forest area.

Success criterion: the model's RMSE must be lower than the zero-change baseline's RMSE on statistical neighbourhoods (Quartiere) not used for training: $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}$. Here, y_i is the measured crown-area change in cell i , and $\hat{y}_i$ is the model's prediction.

Baseline: Predict no change (0 m²) in every cell.

Out of scope: species recommendations and tree-health diagnosis.

3 Originality and Motivation

I am interested in agriculture and in how machine learning can support practical environmental decisions. In my free time, I have built monitoring projects for my plants at home. I have also used Stadt Zürich's Trees Watch (trees.watch), which shows the current tree cadastre. My project would extend that idea by estimating where tree-canopy area may decrease and showing planners where to assess planting needs.

4 Data Sources and Features

Label source: cantonal single-tree datasets for 2017/18 (GIS-ZH 609) and 2021/22 (GIS-ZH 608), accessed through the cantonal OGD WFS as GeoJSON. Attributes include crown area, height, volume, and centroid (LV95); about 157,500 crowns are mapped in Zürich.

Label: for each cell, measured change is the 2021/22 crown-covered area minus the 2017/18 area ($\Delta A = A_{2021/22} - A_{2017/18}$). Calculate area from polygon-cell intersections and dissolve overlapping crowns per cell to avoid double counting.

Limitations: the 2021/22 extraction maps trees at least 3 m high and with crown areas of at least 6 m² (dataset metadata). The two epochs measure net change but do not show when or why it occurred.

Features (start epoch only): crown-covered area per cell, number of detected crowns, and median crown area. Quartiere are used for spatial validation, not as model features. Crown-height features are excluded because height measurements are biased differently between epochs.

Context overlays: Baumersatz (about 352 current replacement sites) and Baumkataster (about 81,000 trees) on the map.

Model inputs: training predictors use start-epoch data only; end-epoch polygons are used only to calculate the measured change. Baumersatz is a map overlay, not a model input.

Validation: test performance on Quartiere not used for training to measure prediction quality in other neighbourhoods during 2017/18–2021/22. Since only one labelled interval is available, accuracy on a later period cannot be assessed until another LiDAR epoch is available.

5 System Design

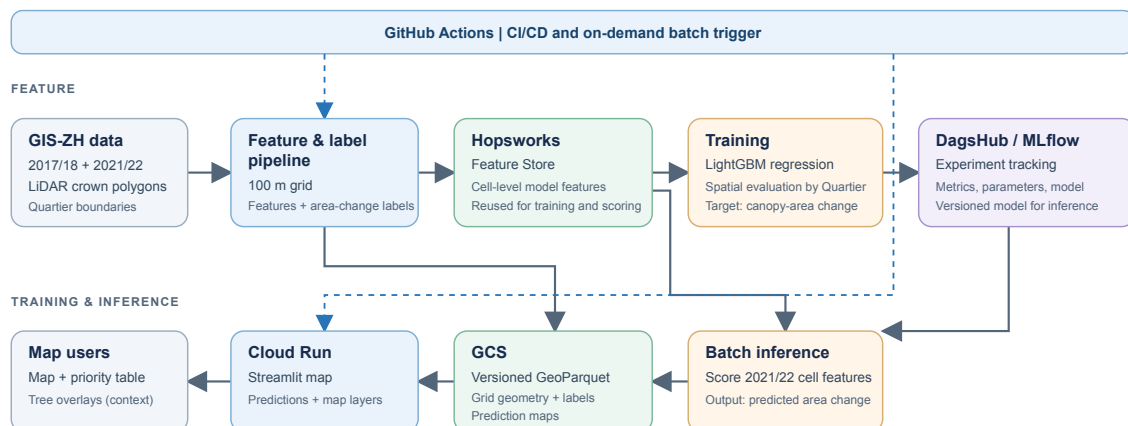


Figure 1: FTI architecture

- **Features and labels:** Process both LiDAR epochs on the same 100 m grid. Store source polygons, grid geometry, labels, and prediction maps as versioned GeoParquet in Google Cloud Storage (GCS). Store cell-level model features in Hopworks.
- **Training and tracking:** Use LightGBM regression for the structured, numeric cell features; its trees can capture non-linear relationships efficiently. Evaluate by Quartier and track runs and model versions with MLflow on DagsHub.
- **Orchestration:** GitHub Actions checks and runs the batch pipeline when a LiDAR epoch becomes available, and tests and deploys the dashboard. No daily schedule is needed.
- **Serving:** A Streamlit app on Google Cloud Run displays batch predictions as map categories, with Baumersatz and Baumkataster as context overlays.
- **Data checks:** Check schema, coordinate system, and spatial coverage during each pipeline run. Reassess prediction accuracy when a new LiDAR epoch is available.